Medicines & Healthcare products Regulatory Agency

CPRD

PART 2: PROTOCOL INFORMATION

Applicants must complete all sections
Study Title Kidney-related adverse events associated with intravitreal anti-vascular endothelial growth factor use in England: a national cohort study
Lay Summary (Max. 250 words) Anti-vascular endothelial growth factor (anti-VEGF) drugs are used to treat eye diseases, and are administered directly into the eyes. Doctors have recently become aware that anti-VEGF administered into the eye might have side effects in other parts of the body. These side effects include high blood pressure, urine protein leak and/or decline in kidney function. Over time, these effects could lead to kidney failure, requiring dialysis or a kidney transplant. However, doctors do not yet know how common or severe these problems might be. We will use electronic health records from GP practices linked to hospital records to determine whether anti-VEGF administered directly into the eye are associated with side effects in other parts of the body in patients with eye disease. We will also assess whether anti-VEGF administered into the eye increases the risk of heart problems and/or death. We think it is important to inform patients about the risks of anti-VEGF treatment (especially kidney failure) as well as the benefits (preventing blindness) to inform decisions about their treatment. With help from our kidney and eye disease patient group, we will quantify these risks and summarise them in a way that is meaningful for patients. We will also use this information to inform doctors about how best to monitor patients who have been treated with anti-VEGF for eye disease.
Technical Summary (Max. 300 words) Anti-vascular endothelial growth factor (anti-VEGF) drugs block the development of new blood vessels and are increasingly used through intravitreal injection to treat age-related macular degeneration, diabetic macular oedema and other eye diseases. There is known to be systemic absorption of anti-VEGF from intravitreal injection, and there are several case-reports of systemic side effects including development of proteinuria and decline in kidney function. Analysis of adverse events from the original clinical trials for ophthalmic indications shows a signal of harm but these trials were small, of limited duration and did not have adjudicated cardio-renal end points. We will use data from the clinical practice research datalink (CPRD) linked to hospital episode statistics (HES) to carry out a cohort study and self-controlled risk interval study to measure the association between anti-VEGF intravitreal injections and kidney-related adverse events. We will use two study populations: 1) people without diabetes who have anti-VEGF intravitreal injections (exposed) versus cataract surgery (comparator), and 2) people with diabetes who have anti-VEGF intravitreal injections (exposed) versus laser photocoagulation (comparator), exposure and comparator status will be determined through OPCS codes in linked HES data. The primary outcome will be ≥30% reduction in estimated glomerular filtration rate (eGFR) determined through blood tests in the primary care record. In the cohort study we will use a propensity score weighted Cox proportional hazards regression model. The self-controlled risk interval approach uses cases only. We will compare the risk of the outcome during the year after first anti-VEGF intravitreal injection to the risk in the year prior, using a conditional Poisson regression model. We will provide definitive information to patients and clinicians about the potential benefits and risks of treatment, to develop adequate monitoring if needed, and to prompt consideration of whether further randomised trials of enhanced vascular prevention during treatment are needed.

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Outcomes to be Measured

Primary outcome: ≥40% reduction in estimated glomerular filtration rate (eGFR)

Secondary outcomes: progression of albuminuria stage (A0 to A1 or 2, or A1 to A2); change in slope of eGFR progression; kidney failure defined as dialysis or kidney transplantation; development of coded nephrotic syndrome; a new diagnosis of hypertension; atrial fibrillation/flutter; myocardial infarction; peripheral arterial ischaemia identified through coded revascularisation procedures or diagnosis of deep vein thrombosis (DVT) or pulmonary embolism (PE); ischaemic or haemorrhagic stroke; heart failure; cardiovascular death; all-cause mortality.

Control outcomes: diagnosis of herpes zoster; hip fracture

Objectives, Specific Aims and Rationale

The aim of this study is to determine whether anti-VEGF intravitreal injections for eye disease are associated with kidney-related adverse events.

The specific objectives are:

1. To estimate whether anti-VEGF drugs are associated with:
    a. accelerated deterioration in kidney function
    b. increased risk of cardiovascular events and mortality
2. To examine whether these outcomes are influenced by level of kidney function and proteinuria, ethnicity, deprivation and sex at the time of initiation of intravitreal anti-VEGF treatment.
3. To explore whether healthcare utilisation, measured through outpatient appointment data, can aid measurement of disease severity.

There have been reports of systemic side effects of anti-VEGF intravitreal injections, but analysis of adverse events in clinical trial data have been limited by small numbers and short duration [1]. Anti-VEGF intravitreal injections are increasingly being used to treat eye disease in patients who are already at increased risk of cardiovascular and renal disease [2]. A better understanding of this association would inform treatment and could aid development of monitoring if necessary.

Study Background

Systemic anti -VEGF drugs are used intravenously to treat cancer. Anti-VEGF drugs are also used to treat retinal diseases via injection into the eye (intravitreal injection).

Systemic anti-VEGF drug use has been associated with hypertension [3], heart failure, myocardial infarction [4], nephropathy and renal failure [4]. It is recommended that patients with pre-existing hypertension, proteinuria, cardiovascular disease and severe renal impairment are monitored. For people receiving anti-VEGF intravitreal injections, there are no monitoring recommendations.

Anti-VEGF intravitreal injections are administered at a lower dose than systemic anti-VEGFs (<5% of the intravenous dose) and it was thought that effects would remain local within the eye [5]. However, there is increasing evidence of systemic absorption. A pharmacokinetic study of 56 patients with age-related macular degeneration, detected ranibizumab, aflibercept and bevacizumab in the circulation, after 3 intravitreal administrations [6]. Ranibizumab was cleared within days, but aflibercept and bevacizumab accumulated over repeated doses. Small studies (<100 patients) have had mixed findings regarding systemic hypertension, with one showing increased risk of hypertension [7], but another finding no change in blood pressure [8]. A recent systematic review and meta-analysis conducted by our team based on 78 trials for anti-VEGF intravitreal injections did not identify an increased risk of cardiorenal outcomes, but showed an increased mortality risk in treated participants with diabetic eye disease [1]. Typically, a treatment course would consist of 3 x monthly injections usually followed by approximately another 4 to 5 injections in year 1. The cardiorenal side effects of anti-VEGF intravitreal injections may not have been determined during initial trials due to relatively short duration of follow-up to see changes in eye disease, while a longer observation period is needed to identify an increased risk of cardiorenal outcomes. Furthermore, it is difficult to assess the true impact of intravitreal anti-VEGF in patients with diabetic maculopathy, who are already at risk of diabetic renal disease, without an appropriate comparison group. One large observational study has been conducted in Taiwan and found an association between Ranibizumab and increased risk of chronic kidney disease. However, this study was only able to investigate one type of anti-VEGF medication and could not adjust for important confounders such as smoking and BMI [9].

Study Type
Descriptive and hypothesis testing

Study Design

We will use two study designs: a cohort study and self-controlled risk interval design to estimate associations between anti-VEGF intravitreal injections and the outcomes to be investigated.

We will use a cohort study design to follow-up people from their first intravitreal anti-VEGF injection or comparator procedure, for renal outcomes. This design facilitates measuring multiple outcomes, and will allow us to measure the incidence, relative risk and absolute risk of outcomes.

Anti-VEGF intravitreal injections are the main treatment for people with age-related macular degeneration, retinal vein occlusion or myopic choroidal neovascularisation. We will therefore compare people receiving anti-VEGF intravitreal injections for these indications to people who have cataracts surgery. People with diabetes are already at higher risk of diabetic renal disease, and have additional options for treatment of diabetic retinopathy, principally laser photocoagulation. Therefore, for people with diabetic retinopathy we will compare anti-VEGF intravitreal injections to laser photocoagulation.

Although we plan to use methods to adjust for confounding, there may be residual confounding. We have tried to identify suitable comparators but we do expect that people receiving laser photocoagulation may have more severe diabetes. Similarly people with cataracts may have different characteristics to people with anti-VEGF indications. Therefore in addition to the cohort study design we will use a self-controlled risk interval design. This design is a simplified version of a self-controlled case series (SCCS) design. In a SCCS design only cases are included and exposed and unexposed periods in the same individuals are compared, this means that confounders are accounted for. The difference between the SCCS and self-controlled risk interval approach is that the risk interval approach compares the exposed period to one shorter unexposed period either before or after exposure and only includes exposed individuals. Whereas in the SCCS the control periods can be both before and after exposure and includes both exposed and unexposed cases [10]. The design requires a number of assumptions to be met: 1) events are recurrent and independent, or unique and uncommon, 2) exposure is not affected by the outcome, 3) the outcome must not censor the observation period [11]. For the primary outcome of ≥40% reduction in eGFR, and all other secondary outcomes, except hospitalisation for stroke, revascularisation procedure and heart failure and cardiovascular death, these assumptions are met so a self-controlled risk interval design is appropriate. We will not use this method for outcomes where the assumptions are not met.

Feasibility counts

As our study population is defined based on linked HES data we cannot directly determine the expected number of patients. A study using HES data [2] identified 388 031 anti-VEGF intravitreal injections April 2014-March 2015. As of November 2022 CPRD Aurum covers approximately 23% of UK population, and 81% of this population are eligible for HES linkage (19% of UK population). Therefore we would expect to identify 73,726 anti-VEGF injections in the first year in Aurum. We will be dividing the population into those with and without diabetes.

Diabetes population:
The study found that 18% anti-VEGF intravitreal injections were for diabetes resulting in 13,271 procedures. If we assume that people receive 8 injections per year this will result in 1,659 people with diabetes. As follow-up would continue over 7 years we would expect new incident cases. For diabetic retinopathy it has been estimated that there are 1 case of severe diabetic retinopathy per 10,000 person-years [5]. In CPRD Aurum there were 15 million people currently registered in November 2022, assuming they remain for the year and 20% were <18 that would result in 12 million, this would result in an estimated 1,200 incident cases per year, potentially resulting in a total study population of around 10,000 people in the diabetes study population.

Non-diabetic population:
If we assume the rest of the population in the study using HES data were a non-diabetic population, this would leave 60,455 procedures. If we assume that people receive 8 injections per year this will result in 7,557 people in the first year. The most frequent indication is age-related macular degeneration (AMD). It is estimated that there are 663 new cases of AMD per million per year [12]. As CPRD Aurum covers 12 million adults, we would expect an additional 7,956 new cases of AMD per year. If we assume half of these have anti-VEGF intravitreal injections, this would result in a study population of over 30,000 people.

Sample size considerations

We used Stata power cox command to estimate the sample size required. If we assume the probability of chronic kidney disease is 5%, based on Chen et al [9], to estimate a hazard ratio of 1.5 at 90% power, we would need N=5114 people. If the event rate was only 2% to estimate the same hazard ratio at the same power we would need a sample size of 12,783 people. The numbers estimated in the feasibility counts section are larger than these numbers, therefore we expect to have the power for the study.

Planned use of linked data (if applicable):

We will use HES admitted patient care data to identify 1) the exposure (intravitreal anti-VEGF injections) and comparator (laser photocoagulation or cataract surgery) status through OPCS procedure codes and 2) hospitalisations for myocardial infarction, peripheral arterial ischaemia identified through coded revascularisation procedures or diagnosis of DVT or PE, ischaemic or haemorrhagic stroke, heart failure.

We will use HES outpatient data to identify appointments with ophthalmology.

We will use ONS death registration data to identify all-cause mortality and cardiovascular mortality specifically.

We will use patient-level 2019 English Index of Multiple Deprivation (IMD) data as a measure of socioeconomic status.

These linkages are essential to the study as the exposure and comparator status will be based on OPCS codes in linked HES admitted patient care data. The output from the study will provide patients and clinicians with definitive information about any additional risk of adverse events associated with anti-VEGF intravitreal injections. In addition, if risks are identified monitoring can be developed and to prompt consideration of whether further randomised trials of enhanced vascular prevention during treatment are needed.

Definition of the Study population

Cohort study:
Study period: 1st March 2013 to 1st March 2020. The study period start date is based on the first NICE approval for the use of anti-VEGF intravitreal injections (ranibizumab) for diabetic macular oedema was 27th February 2013, approval for other indications was already in place at this point. The study period end date is based on the availability HES linkage and the start of the pandemic, HES linkage is available until 31st March 2021 but we felt anti-VEGF intravitreal injections and eGFR measurements would have been impacted by the pandemic, therefore we are ending follow-up on 1st March 2020.

The source population will be people:
- aged 18 years or older
- registered with GP practices meeting CPRD quality control standards
- with at least one year of continuous registration history in CPRD before the first recorded treatment of interest to allow adequate time for recording baseline covariates
- with at least one eGFR measurement in the 18 months prior to index date (index date defined below)
- with linkage to HES

Anti-VEGF intravitreal injections are prescribed for neovascular age-related macular degeneration, retinal vein occlusion, myopic choroidal neovascularisation and diabetic retinopathy. As those with diabetes are likely to have different risks for renal outcomes compared to other groups and have different treatment options, we will have different exposure and comparator groups depending on whether people have diabetes (as described in the study design section), therefore there will be different inclusion and exclusion criteria for each group:

Study population without diabetes:
Exposed inclusion criteria:
- OPCS code for anti-VEGF intravitreal injections (OPCS codes: C79.4, C89.3)

Comparator inclusion criteria:
- OPCS code for cataract surgery (Appendix 1)

Exposed and comparator exclusion criteria:
- Diagnosis of diabetes determined through SNOMED-CT codes in the observation file (Appendix 2)
- Prior eye disease treatment photocoagulation/anti-VEGF/steroid implant/cataract determined through OPCS codes in linked HES APC data.
- MI or stroke in the 6 months prior to index date identified through linked HES APC data and SNOMED-CT codes in the observation file.

Study population with diabetes:
Exposed inclusion criteria:
- OPCS code for anti-VEGF injections
- Diagnosis of diabetes determined through SNOMED-CT codes in the observation file.

Comparator inclusion criteria:
- OPCS code for laser photocoagulation (OPCS codes: C66.4, C82.5, C82.6, C82.5)
- Diagnosis of diabetes determined through SNOMED-CT codes in the observation file.

Exposed and comparator exclusion criteria:
- Prior eye disease treatment photocoagulation/VEGF inhibitor/steroid implant/cataract determined through OPCS codes in the HES APC data.
- MI or stroke in the 6 months prior to index date identified through linked HES APC data and SNOMED-CT codes in the observation file.

The index date will be the first anti-VEGF intravitreal injection, cataract or laser photocoagulation. People will be followed until leaving the practice, death or end of study period, whichever is earliest.

Self-controlled risk interval

The study population for this part of the study will be a subset of the cohort study population. We will identify exposed cases from the cohort study population i.e. all those with each of the outcomes to be used with this design.

The follow-up period for each person will be one year prior to (control period) and one year after (exposed period) the first anti-VEGF intravitreal injection.

Selection of comparison group(s) or controls

As described in the study population section, for the cohort study, the comparison group for those with diabetes will be those with laser photocoagulation determined through OPCS codes in linked HES admitted patient care data. The comparison group for those without diabetes will be those with cataracts surgery determined through OPCS codes linked HES admitted patient care data. Comparators will be followed from the first laser photocoagulation/cataracts code until earliest of leaving the practice, death or end of study period.

For the self-controlled risk interval, exposed and control periods are compared within the same people.

Exposures, Outcomes and Covariates

Exposure:

- Anti-VEGF intravitreal injection determined from linked HES admitted patient care data using OPCS codes (defined above). For the primary analysis a person will be considered exposed from their first injection, analogous to an intention-to-treat analysis in a clinical trial [13]. In a sensitivity analysis we will censor people 6 months after their last anti-VEGF intravitreal injection, analogous to a per-protocol analysis.

Outcomes:

Primary outcome:

- ≥40% reduction in estimated glomerular filtration rate (eGFR) relative to the most recent creatinine measurement prior to index date based on creatinine measurements determined from the observation file.

Secondary outcomes:

- Progression of albuminuria stage relative to the most recent albumin and creatinine measurements on the same day, prior to index date, based on blood test results from observation file.
- Change in slope of eGFR progression based on blood test results from observation file.
- Kidney failure defined as dialysis or kidney transplantation determined using ICD-10 codes in HES APC data and SNOMED-CT codes in the observation file (appendix 3), or eGFR <15ml/min/1.73^2.
- Development of coded nephrotic syndrome determined through SNOMED-CT codes in the observation file (appendix 4) and ICD-10 code N04 in HES admitted patient care data.
- A new diagnosis of hypertension defined as either (i) two consecutive systolic blood pressure (SBP) readings ≥140 mmHg within a year in the observation file, (ii) two consecutive diastolic blood pressure (DBP) readings ≥90 mmHg within a year in the observation file, (iii) a hypertension SNOMED-CT codes in the observation file and antihypertensive medication codes in the drug issue file on at least 2 dates, 6 months either side of the hypertension code (appendix 5).
- Atrial fibrillation/flutter determined using SNOMED-CT codes in the observation file and ICD-10 codes in HES admitted patient care data.
- Myocardial infarction, peripheral arterial ischaemia identified through coded revascularisation procedures or diagnosis of deep vein thrombosis (DVT) or pulmonary embolism (PE), ischaemic or haemorrhagic stroke, heart failure determined using ICD-10 codes in HES admitted patient care data and SNOMED-CT codes in the observation file (appendix 6).
- Cardiovascular death determined using the underlying cause of death in ONS death registry data.

Control outcomes:

- Herpes zoster diagnoses will be based on SNOMED-CT codes in the observation file and ICD-10 codes in HES APC data.
- Hip fractures will be based on SNOMED-CT codes in the observation file.

Covariates

The propensity score model will include confounders:

- **Demographics**: including age, sex, ethnicity determined from the ethnicity record, IMD determined from linked 2019 IMD data, smoking determined using SNOMED-CT codes in the observation file and therapy codes for smoking cessation therapies in the therapy file, body mass index determined through the observation file.
- **Diabetes indicators**: diabetes type (1 vs 2), duration of diabetes at index date, diabetes treatment type at index date (single therapy vs dual therapy vs triple therapy) determined using product codes in the therapy file, duration of diabetic eye disease at index date, amputation or neuropathy through SNOMED-CT codes in the observation file and ICD-10 codes in HES APC data.
- **Comorbidities** at index date including: myocardial infarction, stroke, heart failure, arrhythmia, peripheral vascular disease, cancer, depression, chronic obstructive pulmonary disease, hypertension determined through SNOMED-CT codes in the observation file and ICD-10 codes in HES APC data.
- **Anti-VEGF indicator:** Calendar year of treatment initiation.
- **Medications at index date**: statins, antihypertensives, anti-platelet agents, proton pump inhibitors, loop diuretics, anti-coagulants determined through product codes in the drug issue file.
- **Health care utilisation markers** in the year prior to index date including referral to secondary care determined from the referral file, the number of measures of creatinine to calculate eGFR and proteinuria determined from the observation file, and the number of ophthalmology outpatient appointments determined from HES outpatient data.

Data/ Statistical Analysis

Data preparation:

Ethnicity will be determined from ethnicity record, and categorised as White, Mixed, Black, Asian and Other. Anyone with missing ethnicity will be grouped into a missing category.

IMD will be grouped into quintiles.

eGFR will be continuous and categorised as ≥90ml/min/1.73 m2 = normal and high, 60-89ml/min/1.73 m2 = mild reduction, 45-59ml/min/1.73 m2 = mild-moderate reduction, 30-44ml/min/1.73 m2 moderate to severe reduction, 15-29ml/min/1.73 m2 = severe reduction, <15ml/min/1.73 m2 = kidney failure.

Proteinuria stage will be categorised as albumin:creatinine ratio <3mg/mmol = A1 (normal to mildly increased urine protein levels), 3-30mg/mmol = A2 (moderately increased urine protein levels), >30mg/mmol = A3 (severely increased urine protein levels).

Comorbidities will be defined by having a code for the comorbidity at any point prior to index date.

Medications at index date will be defined as a prescriptions within 30 days of index date.

Descriptive analysis:
We will first describe the characteristics of the exposed and control groups in terms of all covariates. We will describe anti-VEGF intravitreal injection frequency. We will describe the rates of each outcome in each group.

Objective 1: To estimate whether anti-VEGF drugs are associated with: a) accelerated deterioration in kidney function, b) increased risk of cardiovascular events and mortality.

To measure the association between anti-VEGF intravitreal injections and each of the outcomes we will fit a propensity score weighted Cox proportional hazards regression model to estimate hazard ratios. Robust standard errors will be applied. We will estimate propensity scores using multivariate logistic regression models for anti-VEGF injections versus the control (cataracts in non-diabetic group or laser photocoagulation in the diabetic group).

To estimate absolute effects, we will derive cumulative incidence plots using Royston-Parmer models, standardised to the covariate distribution in the exposed groups.

For a) we will also use a self-controlled risk interval design, and use a conditional Poisson regression model to estimate the risk ratio of exposure relative to the control period.

Objective 2: To examine whether these outcomes are influenced by level of kidney function and proteinuria, ethnicity, deprivation and sex at the time of initiation of intravitreal anti-VEGF treatment.

We will stratify the analyses for objective 1 by 1) eGFR category at index date, 2) proteinuria category at index date, 3) ethnicity group, 4) IMD quintile and 5) sex.

Objective 3: To explore whether healthcare utilisation, measured through outpatient appointment data, can aid measurement of disease severity.

We will identify the number of outpatient appointments each year and describe this by eGFR and proteinuria category.

Sensitivity analyses

1. As defined in “Exposures, outcomes and covariates” section, we will censor exposure to anti-VEGF intravitreal injections at 6 months after the last injection if injections stop during follow-up. This is analogous to a per-protocol analysis and aims to reduce misclassification bias if a person is no longer exposed.
2. Primary outcome definition: 1) we will measure a ≥40% reduction in eGFR from the median eGFR at baseline, 2) we will require the ≥40% reduction in eGFR to be sustained over 2 measurements.
3. We will restrict our diabetes sample to those with a SNOMED-CT code for diabetic retinopathy.
4. For the kidney failure outcome, we will restrict our analysis to those with coded dialysis or kidney transplantation only as people with missing eGFR measures may not be random.

5. We will explore whether time since starting exposure is important by stratifying on time [14]: restricting follow-up to 0-6-months, 0-12 months, 0-24 months, 0-36 months, 0-48 months & 0-60 months.
6. We will restrict to those without missing ethnicity.

3

Plan for addressing confounding

As described above we will use propensity score weighting to adjust for confounding in our Cox proportional hazard regression models. We will use control outcomes, where an association seen with control outcomes may indicate unmeasured confounding. We will also use a self-controlled risk interval method which accounts for confounding inherently. Differences in the results may indicate unmeasured confounding in the Cox proportional hazard regression models.

Plans for addressing missing data

We will assess and report the extent of missing data for all variables. We expect there to be missing data for ethnicity and will use a missing category for missing ethnicity and explore the effect of this by restricting to those without missing ethnicity in a sensitivity analysis.

People without an eGFR measurement in 18 months prior to index date will be excluded, as eGFR at baseline is essential to measure the outcome. It has been estimated that kidney function is checked on average ≥1 time per year in those aged over 50 years [15], so would not expect it to have a large impact on the number of people included. We will, however report the number excluded and explore potential bias by comparing the characteristics of those excluded from the study to those included.

Patient or user group involvement

The West of Scotland Kidney Research PPIE patient group has guided the research priorities: quantifying risks of both lack of treatment with anti-VEGF (blindness) with risks of receiving treatment (which may include kidney failure, heart disease and early death). PPIE input has helped identify the best methods to quantify these risks for patients, i.e., summarising the time frames in which they may experience benefits and risks of anti-VEGF treatments, rather than just quantifying these as percentage risks within a time frame. Our PPIE team have agreed to continue to contribute to the work of this study, including helping with generating sensitive, readily understood and well contextualised patient information for dissemination of our results.

Plans for disseminating and communicating study results

We will publish the results in a peer reviewed journal, and scientific conferences such as UK kidney week and the International Society for Pharmacoepidemiology.

Conflict of interest statement

REC has personal shares in AstraZeneca. LT has grant funding from MRC, the Wellcome Trust and is funded by an NIHR Research Professorship (NIHR302405), has consulted for Bayer and is on the MHRA expert advisory group (Women's health) and a member of 4 non-industry funded trial advisory committees (unpaid). IJD has grants from GSK and AstraZeneca and holds shares in GSK.

Limitations of the study design, data sources, and analytic methods

Identifying VEGF intravitreal injections: we have to rely on OPCS codes to identify these injections, as high-cost drugs prescribing data is unavailable. OPCS codes are necessary for hospital reimbursement so we hope we would capture most exposure, but it is possible some people will be misclassified. In addition, as we are only using procedure codes we will not be able to distinguish between different types of anti-VEGF drugs used, therefore we will not be able to estimate whether risk of renal outcomes vary by the type of drug.

Residual confounding: Although we have tried to identify confounders, there is the possibility that there will be residual confounding in the Cox proportional hazards models. However we are also using a self-controlled risk interval method for the renal outcomes, so can assess this.

The main outcome of eGFR relies on creatinine blood test results in the primary care record, it is possible that there may be measures that take place at hospitals and are not captured within the primary care record, this is likely to vary by region. We will adjust for region within our models.

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List of Appendices Appendix 1: Provisional codelist for cataract surgery (OPCS) Appendix 2: Provisional codelist for diabetes (SNOMED-CT) Appendix 3: Provisional codelist for renal replacement therapy (SNOMED-CT & ICD-10) Appendix 4: Provisional codelist for nephrotic syndrome (SNOMED-CT) Appendix 5: Provisional codelist for hypertension & antihypertensives (SNOMED-CT & Prodcodes) Appendix 6: Provisional codelists for cardiovascular outcomes
Grant ID *(optional)*